

Mathematics

A Cultural Heritage

Lecture 2

Numbers – Part 3

Positional Number Systems

Positional Number Systems

- Same symbols repeating with larger powers
- Roman not positional. Babylonian is (base 60).
- Base 10: 10's place: multiply by 10, 100's place: multiply by 100, etc.

$$2506 = 2 \times 10^3 + 5 \times 10^2 + 0 \times 10^1 + 6 \times 10^0$$

- Binary:

$$101 = 1 \times 4 + 0 \times 2 + 1 \times 1$$

Positional Number System

... the ingenious method of expressing all numbers by means of ten symbols, each symbol receiving a value of position as well as an absolute value... appears so simple to us now that we ignore its true merits. But its very simplicity...puts our arithmetic in the first rank of useful inventions ...remember it escaped the genius of Archimedes and Apollonius...

–Pierre-Simon Laplace



Laplace (1749-1827)

The Number Zero

- If no “10’s”, then you could just leave a space there:
 $3\ 4 = 304$

- What if you got sloppy?
 $34 = 34$ or 304 ?

- Maybe blocks?

3		4
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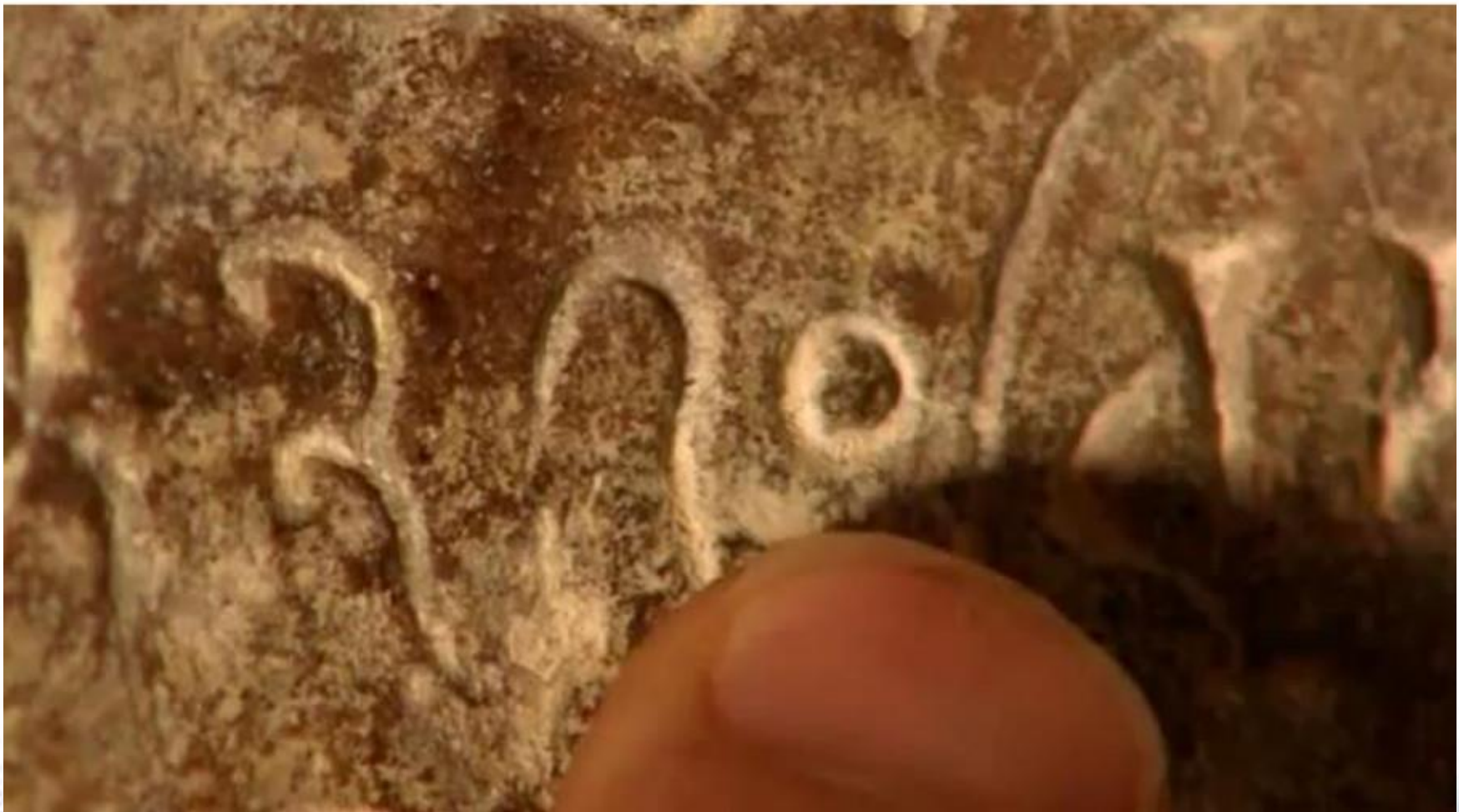
- What if missing spaces are at the end?
- Solution: introduce 0

The Number Zero

- Mayans had a zero (dating at least to 36BC).
- Indians introduced to our system.
- Oldest recorded use: 458AD
- First used in European text by Venetian mathematician Luca Pocioli (1445-1514).

The Number Zero

1. Video Excerpt from “[The Story of 1](#)” (29:00-33:20)



Zero

- Acceptance of **zero** as **nothing** or **number** appears late in history
- Reading Assignment:

A Brief History of Zero by Kristen McQuillin.

www.mediatinker.com/blog/archives/008821.html

Number Bases

- Base 2 (binary): 0,1, Base 16: (0-9, A,B,C,D,E,F)
- Base 3: 0,1,2
- Base 5: 0,1,2,3,4
- Base 10
- Base 60
- Bases 6,12,20
- Mixed-Base systems (years, months, days, ...)
Stardate (Star Trek): 41423.6

Numbers in Different Bases

Decimal	1	2	3	4	5	6	7
Base 2	1	10	11	100	101	110	111
Base 3	1	2	10	11	12	20	21
Base 4	1	2	3	10	11	12	13
Base 5	1	2	3	4	10	11	12
Base 6	1	2	3	4	5	10	11
Base 7	1	2	3	4	5	6	10
Base 2,3,2	1	1,0	1,1	2,0	2,1	1,0,0	1,0,1

Computer Counting

- Binary (why?)
- Shorthand: Bases 8 and 16

Decimal	9	10	11	12	13	14	15
Hexadecimal	9	A	B	C	D	E	F
Decimal	16	17	18	19	20	21	22
Hexadecimal	10	11	12	13	14	15	16
Decimal	23	24	25	26	27	28	29
Hexadecimal	17	18	19	1A	1B	1C	1D
Decimal	30	31	32	33	34	35	36
Hexadecimal	1E	1F	20	21	22	23	24

What Would Mickey Mouse's Number System Be?



Count Their Fingers



Looney Tunes characters